# Did Cooling Measures Really Cool Down Housing Market in Hong Kong?

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10 May, 2025

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**Abstract**

In this article, the study examines the effect of Hong Kong's cooling measures on its overheated housing market. The study provides correlation analysis and case studies for each type of policy category. The effect of each round of policies is estimated through a regression analysis. Demand and supply analysis is included as a theoretical framework. The results show that the cooling measures were successful in slowing the housing price increase, while vary in different manners by house types as well as by policy rounds.

## 1 Background and Introduction

Hong Kong's housing demand is driven by twin needs, accommodation need and investment return need. With its comparatively relaxed capital flow regulation as an open economy, it is susceptible to speculative need (Li & Cheung, 2016). Since the global financial crisis in 2008, Hong Kong has seen an increasingly rising housing price, which fueled social tensions. The government has tried in response by levying more than 10 rounds of cooling measures since October 2009 (Table 3), including differentiated stamps duties and tightened mortgage lending conditions, in an effort to cool the overheating housing market. The COVID-19 outbreak in 2020 has, however, inflicted a heavy blow on Hong Kong's economy and pushed the government to relax some of these cooling measures in 2023. (Deng et al., 2024).

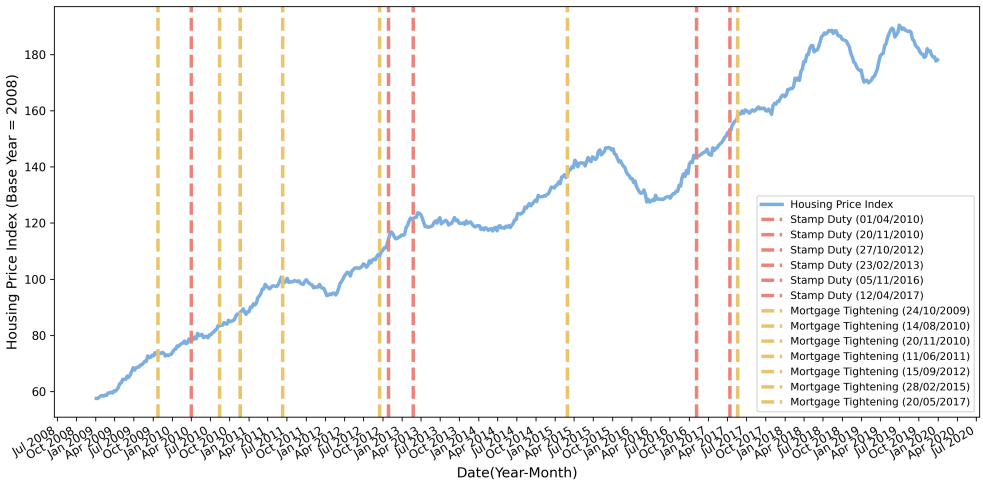


Figure 1: Housing Price Index with Cooling Measures

This graph illustrates the the house price index (the Centa-City Leading Index, CCL) alongside the cooling measures timeline from July 2008 to July 2020. Overall, it reveals an upward trend in housing prices, punctuated by short-term dips following the implementation of cooling measures. *Data source: Centa Data*

To briefly introduce the cooling measures, stamp duties consist of Special Stamp Duty (SSD), Buyer's Stamp Duty (BSD), and the Ad Valorem Stamp Duty (AVD). Our analysis of this type of policy mainly focus on BSD, which applies to all residential property transactions except for Hong Kong Permanent Residents (HKPRs). Corporate entities and non-HKPR individuals must pay BSD at 15% (from January 2012 to October 26, 2023) and 7.5% (on or after October 26, 2023)(Inland Revenue Department, 2024). Mortgage-tightening measures primarily involve tightened loan-to-value (LTV) ratio caps and stricter debt service ratio (DSR) limits. LTV ratio is the ratio of the maximum loan amount to property value, and DSR is the ratio of annual debt repayment amount to annual income.

# 2 Literature Review

Since the Asian housing boom after the Global Financial Crisis, there has been academic research focusing on both the housing market and the role of prudent policies in Asia. However, academic research assessing the effectiveness of the cooling measures is limited.

Most existing studies focused on transaction-level data. Using such data, Ho et al. (2018) examined the effects of selected cooling measures (mortgage tightening policies and stamp duty) on Hong Kong's secondary housing market. However, their analysis relied on a relatively small sample of fewer than 30,000 transactions from 1997 to 2016 and was limited to the secondary market, which introduced several empirical limitations. Moreover, most of the literature emphasized short-term impacts. For example, even adding the methodology

of Regression discontinuity design (RDD), Deng et al. (2024) only assessed the immediate effects of cooling measures without considering the long-term influence of different policy rounds.

Internationally, similar measures were implemented. For example, Deng et al. (2019) proved that the stamp duty in Singapore between 2009 and 2013 had led to a 10%–15% decline in their house prices, without significant disruption of the economy. Similarly, Krznar and Morsink (2014) found that the mortgage tightening measures in Canada helped slow the rise in house prices during the period from 1998 to 2013. However, the specific effect of housing market cooling measures in Hong Kong remained inconclusive.

While previous studies provided useful insights, they often focused on a narrow set of policies and only emphasized short-term effects. To fill this gap, our study aims to conduct a more comprehensive analysis of various policy types and examine their long-term impacts of each policy round, thereby assessing the effectiveness of Hong Kong's cooling measures in moderating house price growth.

# 3 Methodology

To measure housing prices in Hong Kong, the weekly data Centa-City Leading Index (CCL) was chosen to represent the overall housing price. We also included housing prices across five different property sizes[1] from the Rating and Valuation Department of Hong Kong. The exact BSD rate and LTV ratio restrictions were extracted from the Inland Revenue Department of Hong Kong and Hong Kong Monetary Authority. Quarterly GDP from Cencus and Statistics Department was also used for analysis.

Correlation Analysis was conducted by comparing the housing price growth rate under different BSD rates and LTV ratio restrictions. Case studies of houses with certain sizes were included. The analysis concerning BSD involved CCL from 2009 to 2024, to include three different BSD rates. The monthly growth rates of CCL and housing prices categorized by five different house sizes (A-E), were calculated and presented using box plots to find out the relationships. The analysis with LTV ratio included housing prices from 2009 to 2019, to avoid market disruptions from COVID-19 pandemic. Since the LTV ratio restrictions were put based on property value without general standards, and the housing price data were only categorized by sizes, we estimated the property values of houses in different size classes by the formula below and compared the calculated housing prices growth rates under different LTV ratios.

$$P_{\mathrm{i}} = \overline{A}_i \times P_{\mathrm{avg}}$$

- $P_{\text{Class i}}$: Estimated housing price of houses of Class i (A-E) at a given time
- $\overline{A}_i$: Mean square meter of houses from Class i

[1] Class A: below 40m$^2$; Class B: 40m$^2$–69.9m$^2$; Class C: 70m$^2$–99.9m; Class D: 100m$^2$–159.9m$^2$; Class E: 160m$^2$ or above

- $P_{\text{avg}}$:Average housing price per square meter at a given time

A time-seires regression regressing CCL was conducted to assess the impact of each policy round. The sample spanned from 1994 to 2019, constructing a lengthy pre-policy baseline, while avoiding distortions from the COVID-19. Considering the underlying upward market trend and broad economic conditions, a linear time trend (t) and GDP were included as control variables. We created dummy variables for each policy round, including three categories:

1. $A_1$-$A_7$: Seven rounds of LTV ratio restrictions
2. $B_1$-$B_4$: Four rounds of DSR restrictions
3. $C$: stamp duty introduction

To simplify the analysis, we only considered the launch of stamp duty. Since different stamp duty rounds applied to specific transaction types, evaluating their impact across the entire housing market was complex and may introduce variability. Each policy dummy was activated when a specific measure took effect and deactivated once the next round of that policy type was introduced. Therefore, only one version of each policy was active at any given time. This ensured that policy dummies do not overlap within their category. We then interacted each policy dummy with the time trend.

The regression model is specified as:

$$CCL_t = \beta_0 + \beta_1 t + \beta_2 \text{GDP}_t + \sum_{i=1}^{7} [\beta_A A_i + \beta_{A,T}(t \times A_i)] + \sum_{j=1}^{4} [\beta_B B_j + \beta_{B,T}(t \times B_j)] + \beta_C C + \beta_{C,T}(t \times C) + \epsilon_t$$

The interaction terms ($t \times A_i$, $t \times B_j$, $t \times C$) estimated the impact of each policy on the incremental change in slope, allowing us to assess how these measures altered the trajectory of housing price growth.

Demand and supply analysis was used to explain how these policies affect the housing market, exploring the underlying mechanics of the effectiveness of the cooling measures.

# 4 Findings

## 4.1 Correlation Analysis

### 4.1.1 BSD and Housing Price Growth Rate

Figure 2 demonstrates an inverse correlation between the two variables. As the BSD rate increases from 0% to 7.5% to 15%, there is a significant downward shift in the distribution of CCL growth rates. Specifically, the mean of CCL growth rate transits from the positive

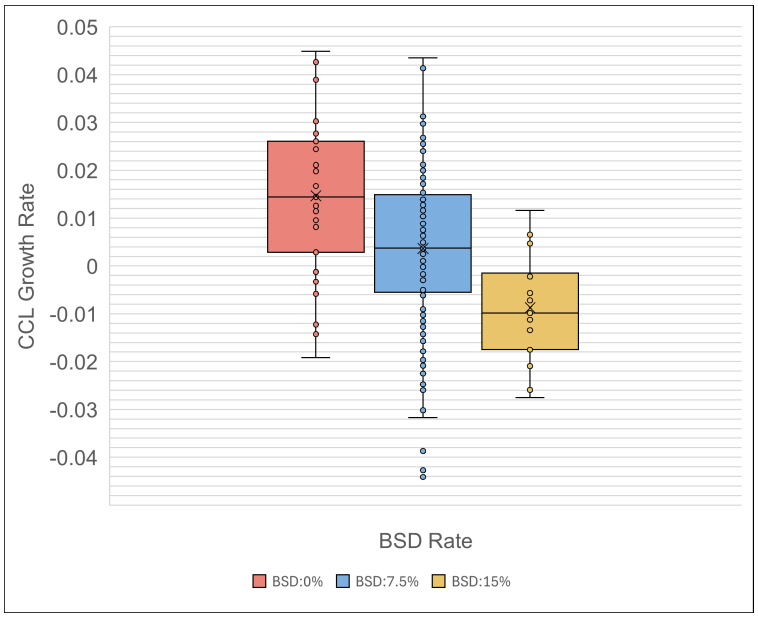


Figure 2: BSD and Overall CCL Growth Rate

*Data source: Centa Data; Inland Revenue Department*

zone to the negative zone. Additionally, more points cluster near zero. These trends indicate that higher BSD rates effectively slow and even reverse the overall house price growth.

The statistical analysis confirms these visual observations. As shown in Table 1, by regressing the CCL growth rate, the coefficient of BSD rate is negative, and the p-value of the t-test hovers around the 0.05 significance threshold.

| **Variable** | **Coefficient** | **P-value** |
|---|---|---|
| Intercept | 0.035478 | 0.00365 |
| BSD Rate (X Variable 1) | -0.17844* | 0.050369 |

Table 1: Regression Output for BSD Rate and Overall CCL Growth Rate

*Data source: Centa Data; Inland Revenue Department*

**Case Study:** The effects of BSD vary on different-size properties. Figure 3 - Figure 7 show the relationship between BSD rate and housing price growth rate for five categories. By comparison, it is idetifiable that BSD has the largest impact on the smallest and the largest units (Class A and Class E), with the most obvious reduction of the mean. This suggests that BSD curbs the rising housing prices of extreme cases (smallest houses and largest houses) more effectively, which could be explained by the fact that small houses attract first-time buyers and large houses appeal to wealthy investors, leading to more speculation focus. Interestingly, the growth rate of the prices in Class C even increases. This suggests that the stamp duty intervention is mostly irrelevant, or incompetent for these houses.

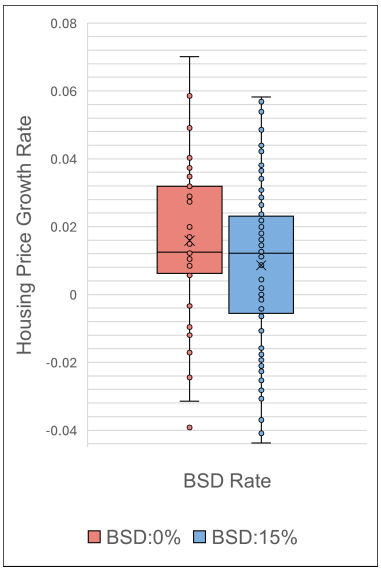


Figure 3: BSD Rate and Housing Price Growth Rate for Houses in Class A (House sizes On or below 40 m$^2$)

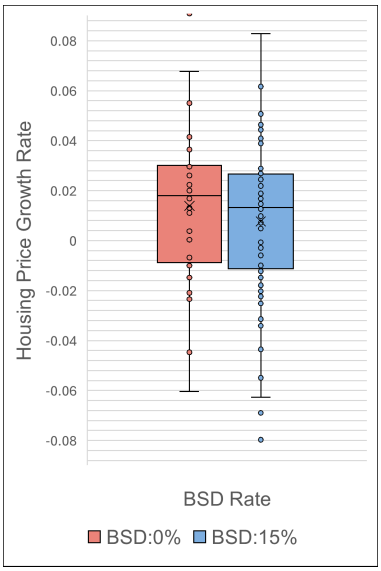


Figure 4: BSD Rate and Housing Price Growth Rate for Houses in Class B (House sizes from 40m$^2$ to 70m$^2$)

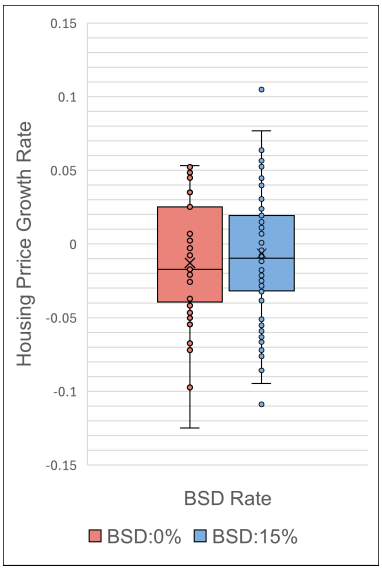


Figure 5: BSD Rate and Housing Price Growth Rate for Houses in Class C (House sizes from 70m$^2$ to 100m$^2$)

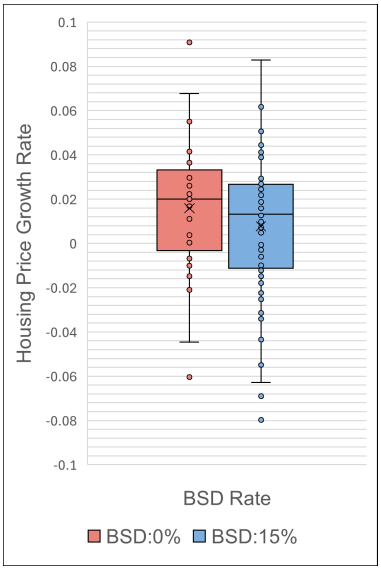


Figure 6: BSD Rate and Housing Price Growth Rate for Houses in Class D (House sizes from 100m$^2$ to 160m$^2$)

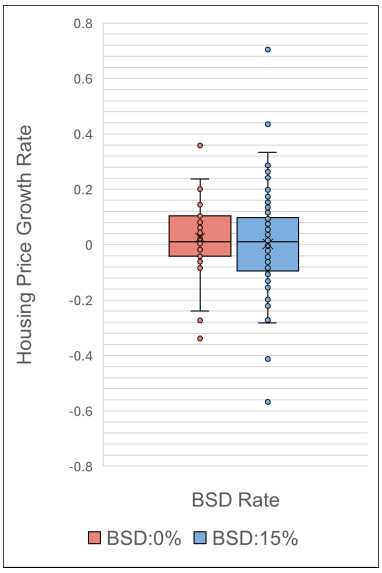


Figure 7: BSD Rate and Housing Price Growth Rate for Houses in Class E (House sizes On or above 160m$^2$)

*Data source: Rating and Valuation Department; Inland Revenue Department*

### 4.1.2 LTV Ratio and Housing Price Growth Rate

As the LTV ratio is adjusted according to different property price segments (Table 4), using the overall housing prices to examine the relationship is neither accurate nor realistic. But through calculation introduced in detail in the Methodology section, we found that the houses larger than 160m$^2$ are mostly priced above 20 million HKD. The most expensive houses were also faced with stricter restrictions. We therefore conducted a case study focusing on properties over 20 million HKD with sizes above 160m$^2$.

**Case Study:** As is shown in Figure 8, when the LTV ratio decreases from 70% to 50%, tightening the policy, the housing price growth rate also has a tendency to decline. Notably, the mean value of the housing price growth rate gradually approaches to zero. Lowering the LTV ratio shows an effectiveness of slowing housing price growth for the most expensive houses.

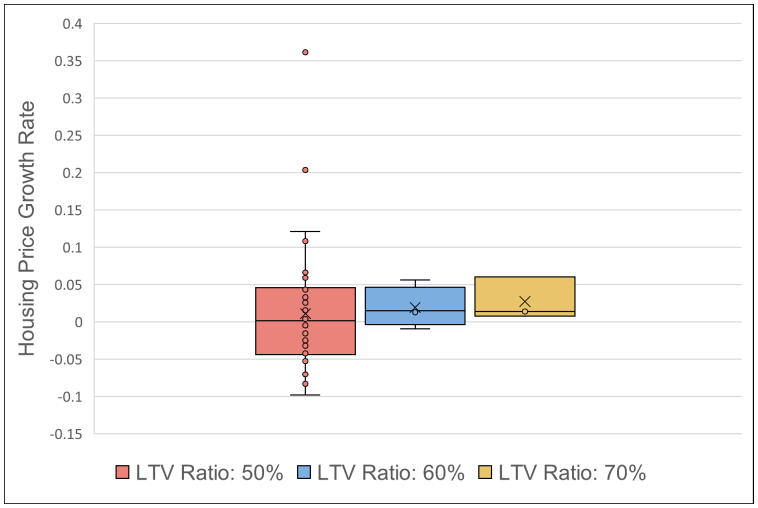


Figure 8: LTV Ratio and Housing Price Growth Rate for Houses larger than 160m$^2$

*Data source: Rating and Valuation Department; Hong Kong Monetary Authority*

## 4.2 Regression Analysis

The regression results show that most policy interventions had a significant decelerating effect on housing prices. Nearly every time-trend interaction coefficient for the policy dummies are negative and statistically significant, meaning that after a tightening measure was introduced, the growth of housing price slowed down compared to the prior trend.

The magnitude of the policy impact, however, was not uniform across all rounds. The interventions in the mid-phase exhibited the strongest effects. $A_5$ and $A_6$, $B_2$ and $B_3$, showed the largest and most significant slope reductions in their own policy categories. This may suggest an accumulative impact due to the sequential interventions. In contrast, late rounds

of policy tightening showed diminishing marginal effects on the housing price trend, with smaller magnitude and lower significance level.

| **Variable** | **Coefficient** | **P-value** |
|---|---|---|
| Intercept | -62.0373 | 0.000*** |
| Time Trend | -0.0873 | 0.000*** |
| GDP | 0.8836 | 0.000*** |
| A1 (Time-Adj.) | -0.4986 | 0.057* |
| A2 (Time-Adj.) | 0.1487 | 0.712 |
| A3 (Time-Adj.) | -0.3247 | 0.121 |
| A4 (Time-Adj.) | -0.3317 | 0.055* |
| A5 (Time-Adj.) | -0.4219 | 0.035** |
| A6 (Time-Adj.) | -0.4508 | 0.025** |
| A7 (Time-Adj.) | -0.3346 | 0.095* |
| B1 (Time-Adj.) | -0.5078 | 0.124 |
| B2 (Time-Adj.) | -0.4219 | 0.035** |
| B3 (Time-Adj.) | -0.4508 | 0.025** |
| B4 (Time-Adj.) | -0.3346 | 0.095* |
| C1 (Time-Adj.) | 0.7900 | 0.048** |
| **No. of Observations** | 1357 | |
| **R-squared** | 0.968 | |

Table 2: Regression Results for CCL with Time-trend Interacted Policy Dummy Vairables

***Note:*** ****: highly significant, **: very significant, *: significant*

## 4.3 Demand and Supply Analysis

The empirical findings above can be explained by the demand and supply framework for the housing market. These cooling measures were primarily demand-side interventions.

The mortgage tightening policies, by limiting leverage, constrain borrowers' ability to finance home purchases. This shift the demand curve downward, and when all else equal,

the market settles at a lower equilibrium price (Figure 9.)

Stamp duty is similar to an ad valorem tax imposed on property transactions, which is proportional to the value of the property purchased. The tax burden grows with the sale price. This causes the demand curve to rotate downward, which also produced a lower equilibrium price.

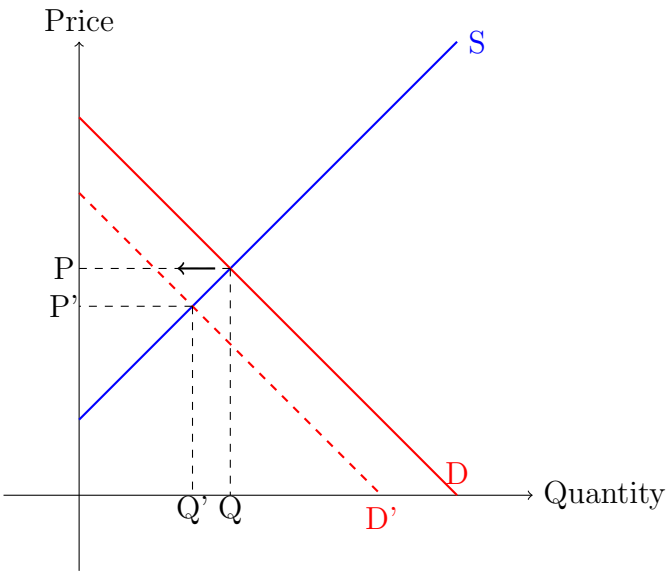


Figure 9: Effects of Mortgage Tightening Policies

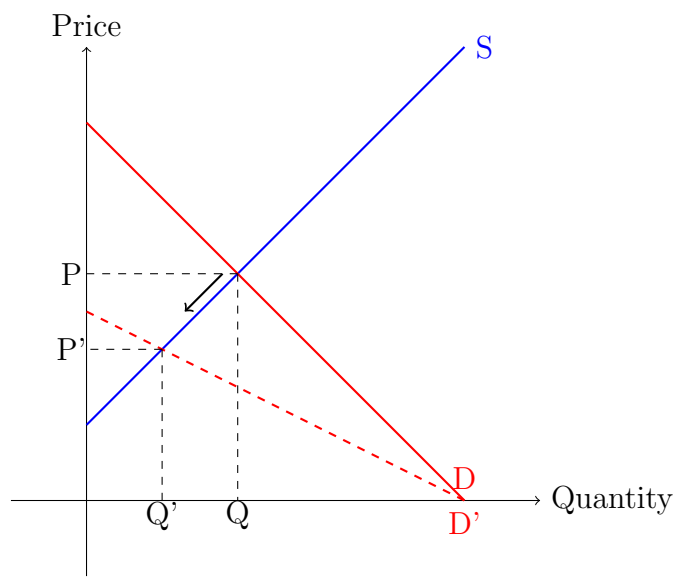


Figure 10: Effects of Stamp Duty Policies

# 5 Conclusion

In conclusion, our analysis showed that the series of cooling measures, both mortgage-tightening and stamp-duty policies, produced a downward effect on housing prices. Additionally, by correlation analysis, we found the effectiveness of these policies vary among different house types; by regression analysis, we found that their impact was accumulative across successive policy rounds. Our study provides some insights for policymaking in the housing market, however, it may not accounted for all influencing factors of housing price, such as transaction volume and policies targeting at the supply side. There is still a long way to further the study.

# References


[1] Census and Statistics Department of Hong Kong SAR. (2025). *Gross Domestic Product (GDP), implicit price deflator of GDP and per capita GDP* [Data set]. https://www.censtatd.gov.hk/en/web_table.html?id=31

[2] Centaline Property Agency Limited. (2025). *Centa-City Leading Index* [Data set]. https://hk.centanet.com/CCI/query/data?type=CCL

[3] Deng, Y., Gyourko, J., & Li, T. (2019). Singapore's cooling measures and its housing market. *Journal of Housing Economics, 45*, 101573. https://doi.org/10.1016/j.jhe.2018.04.001

[4] Deng, Y., Han, C., Li, T., & Wang, Y. (2024). The effectiveness and consequences of the government's interventions for Hong Kong's residential housing markets. *Real Estate Economics, 52*(2), 324–365. https://doi.org/10.1111/1540-6229.12472

[5] Ho, L. S., Wei, X., Wong, G. W. C., & Hu, M. (2018). The impact of cooling measures: Evidence from Hong Kong residential resale market [Unpublished manuscript].

[6] Li, L. H., & Cheung, K. S. (2016). Housing price and transaction intensity correlation in Hong Kong: Implications for government housing policy. *Journal of Housing and the Built Environment, 32*(2), 269–287. https://doi.org/10.1007/s10901-016-9512-7

[7] Krznar, I., & Morsink, J. (2014). With great power comes great responsibility: Macroprudential tools at work in Canada. *IMF Working Papers, 14*(83), 1. https://doi.org/10.5089/9781484383445.001

[8] Rating and Valuation Department of Hong Kong SAR. (2025). *Private Domestic - Average Prices by Class (from 1982)* [Data set]. https://www.rvd.gov.hk/en/publications/property_market_statistics.html

# A Appendix

| Round | Effective Date | Biref Highlightings of Measures |
|---|---|---|
| 1st | 23 Oct, 2009 | Max. LTV ratio of 60% if property value  20 mn HKD<br>Max. LTV ratio of 70% if property value < 20 mn HKD and restricted LTV cap to 12 mn HKD. |
| 2nd | 1 April , 2010 | Stamp duty raised to 4.25% from 3.75% if property value > 20 mn HKD |
| 3rd | 14 Aug 2010 | Max. LTV ratio of 60% if property value  12mn HKD<br>Max. LTV ratio of 70% if property value <12 mn HKD (loan amount capped at 7.2 mn HKD)<br>Standardize DSR limit at 50%. |
| 4th | 20 Nov, 2010 | Max. LTV ratio of 50% if property value 12 mn<br>Max. LTV ratio of 60% if 8 mn HKD  property value ¡12 mn HKD (loan amount capped at 6 mn HKD)<br>Max. LTV ratio of 70% if property value < 8 mn HKD.<br>Max. LTV ratio to 50% for non-owner-occupied properties;<br>(ii) properties held under company names;<br>(iii) commercial and industrial properties;<br>(iv) properties with mortgage loans based on borrower's net asset value. |
| 5th | 11 Jun, 2011 | For Owner-Occupied Property Mortgages:<br>(i)Max. LTV ratio of 50% if 10 mn HKD  property value <12 mn HKD<br>(ii) Max. LTV ratio of 60% if 7 mn HKD  property value <10 mn HKD (loan amount capped at 5 mn HKD)<br>(iii)Max. LTV ratio of 70% if property value <7 mn HKD (loan amount capped at 4.2 mn HKD)<br>For Residential and Non-Residential Property Mortgages:<br>If the borrower's primary income is not from Hong Kong,<br>LTV is reduced by 10%.<br>For asset-based mortgages, LTV is reduced to 40%. |
| 6th | 15 Sep, 2012 | If a mortgage loan applicant owns or guarantees outstanding mortgages:<br>(i) The applicant's DSR is reduced to 40%<br>(ii) If the applicant applies for a new mortgage loan based on asset level, LTV is reduced to 30%.<br>(iii) If the applicant's primary income is not from Hong Kong, LTV is further reduced by 10% points. |
| 7th | 27 Oct, 2012 | BSD at a flat rate of 15% for all residential property acquisitions (HKPR exempted)<br>(ii) LTV for non-residential property mortgages is reduced by 10%.<br>(iii) LTV and DSR for car park mortgages must follow the applicable levels for non-residential property mortgages |
| 8th | 23 Feb, 2013 | AVD raised to 1.5% from 100 HKD if property value <2 mn HKD |
| 9th | 28 Feb, 2015 | Max. LTV ratio of 60% if property value 7 mn HKD(loan amount capped at 6 mn HKD)<br>For a second self-occupied property and non-owner-occupied properties, DSR is reduced to 40%. |
| 10th | 5 Nov, 2016 | AVD jumps to a flat rate of 15% for all residential property acquisitions<br>(non-homeowner HKPR are exempted) |
| 11th | 12 April, 2017 | No exemption from the 15% AVD for HKPRs' acquisition of more than one residential property under a single instrument |
| 12th | 20 May, 2017 | LTV cap reduced by 10 percentage points for mortgages extended to borrowers<br>with one or more pre-existing mortgages<br>DSR limit reduced by 10 percentage points for mortgages extended to borrowers<br>whose income is mainly derived from outside Hong Kong |

Table 3: Timeline of cooling measures

*Data Source: Hong Kong Monetary Authority; Inland Revenue Department of Hong Kong*

| **Date** | **House Price Segment** | | | | |
|---|---|---|---|---|---|
| | **Below 7M** | **7-10M** | **10-12M** | **12-20M** | **Above 20M** |
| 2009/1/1 | 70% | 70% | 70% | 70% | 70% |
| 2009/10/23 | 70% | 70% | 70% | 70% | 60% |
| 2010/8/13 | 70% | 70% | 70% | 60% | 60% |
| 2010/11/19 | 70% | 60% | 60% | 50% | 50% |
| 2011/6/10 | 70% | 60% | 50% | 50% | 50% |
| 2015/2/27 | 70% | 60% | 50% | 50% | 50% |

Table 4: LTV Ratio Restrictions with House Price Segments

*Data Source: Hong Kong Monetary Authority*